Dragon Capsule Aerothermal Re-entry Analysis

Coupled CFD and Thermal Protection System Sizing — Technical Report

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Vehicle	Dragon-style blunt-body capsule, 3.6 m diameter
Software	ANSYS SpaceClaim / Meshing / Fluent 2026 R1 (Student), Python 3
Scope	Mach sweep CFD validated against analytical theory, coupled to transient TPS sizing
Status	Complete

1. Objective

A re-entering capsule dissipates its orbital kinetic energy into a shock layer that can exceed 10,000 K. Sizing a thermal protection system that survives it needs two things: a trustworthy pressure field, and a heating history along the actual trajectory.

Phase 1 runs steady density-based CFD across a Mach sweep and validates it against normal-shock and Newtonian theory rather than assuming the solver is right. **Phase 2** takes stagnation heating from the Fay-Riddell correlation — anchored to the CFD-validated range — pushes it along an Allen-Eggers ballistic entry, and drives a transient conduction model to size ablator thickness against a bondline limit.

> **Why validate against theory at all?** A blunt-body stagnation region is one of the few hypersonic problems with reliable closed-form answers. Rayleigh Pitot and modified Newtonian theory give an independent check the CFD must pass before any of its output is used downstream.

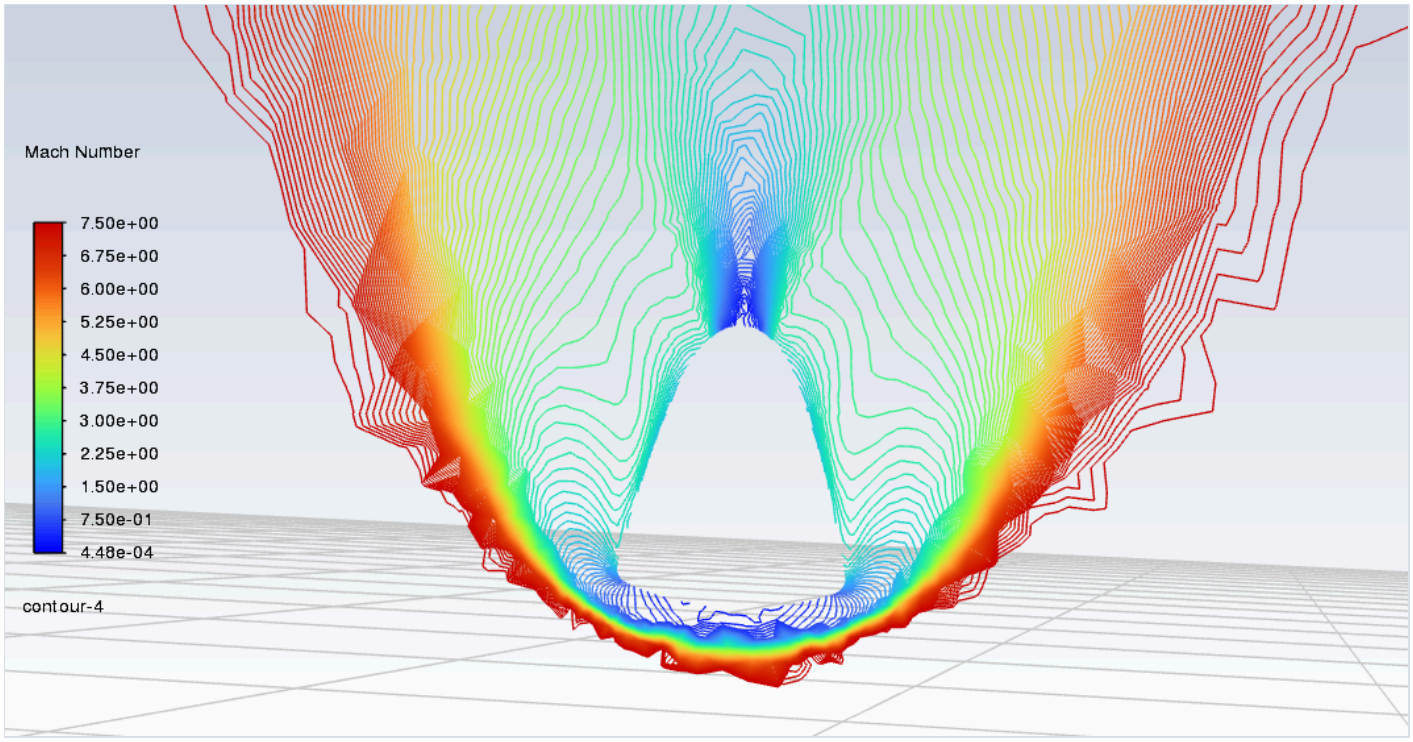


Figure 1 — Mach number contours on the symmetry plane, Mach 7.5. Flow decelerates to subsonic across the detached bow shock, forming the high-pressure, high-temperature shock layer that drives heat-shield sizing.

2. Geometry and domain

Parameter	Value	Parameter	Value
Capsule diameter	3.6 m	Domain radius	≈ 35.6 m (20 × dia)
Fitted nose radius R_n	3.248 m	Domain cushion	700 %
Freestream direction	+Y (0, 1, 0)	Fluid volume	1.893×10^{14} mm³

Two import failures worth recording. SLDPRT import failed outright; STEP succeeded and avoided facet-to-solid conversion. Manual Boolean subtract then failed *silently* — the imported capsule sat off-origin, so the sphere and capsule never intersected and Combine returned the target unchanged with no error. It was caught only because the domain volume exactly matched a bare sphere. The Enclosure tool creates the domain and subtracts the body in one operation regardless of body position.

3. Mesh and grid convergence

479,475 cells, sized deliberately against the 512,000-cell ANSYS Student ceiling. Element size 6.5 m, max 8.5 m, curvature normal angle 10°, no inflation layers.

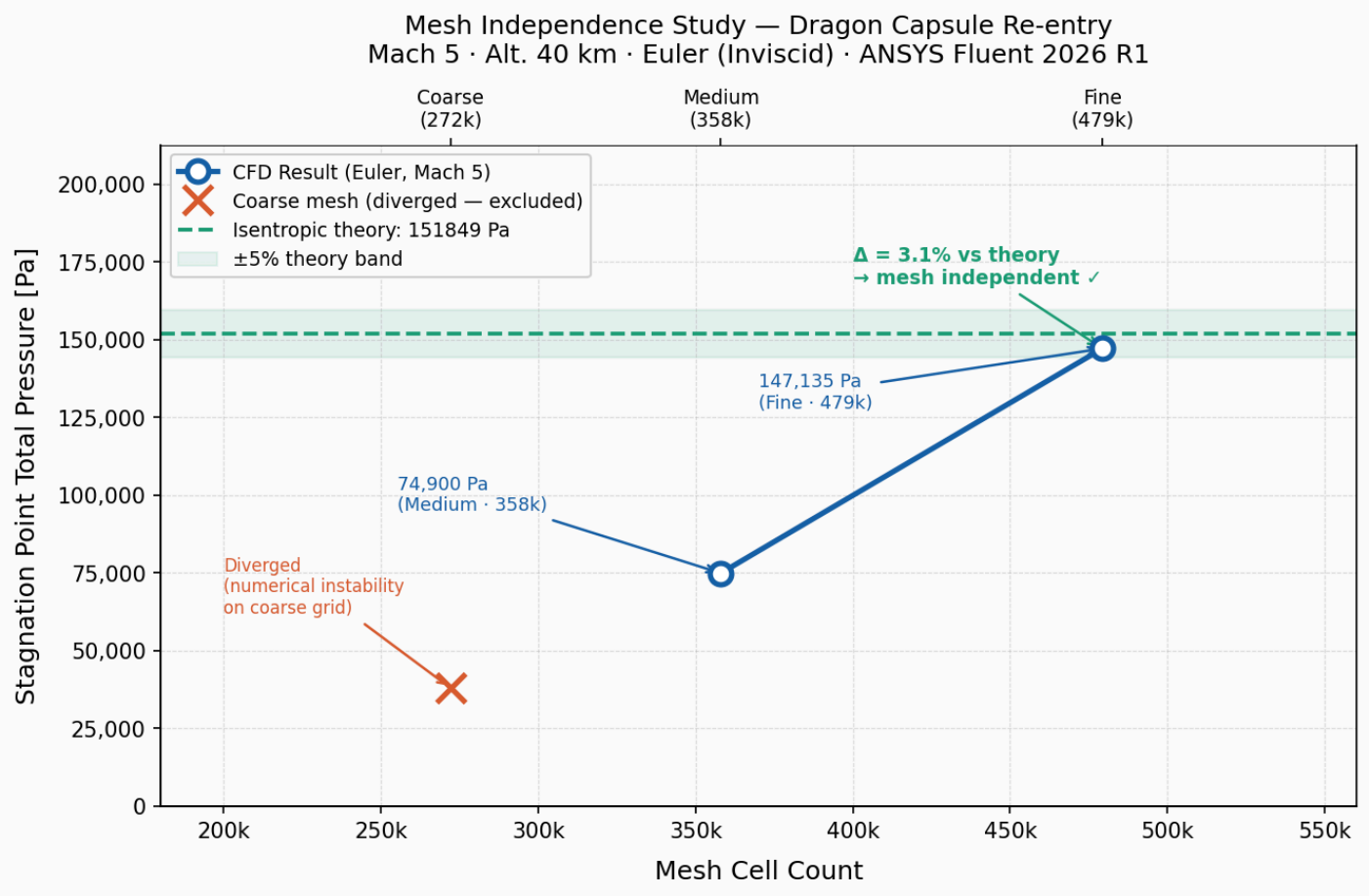


Figure 2 — Grid convergence at Mach 5. The coarse 272k grid diverged; the medium 358k grid under-predicted stagnation pressure by half. The fine 479k grid lands within 3.1% of isentropic theory, inside the ±5% band.

Element size had to be re-tuned from 3.5 m to 6.5 m after the domain was enlarged to 20× diameter — at constant element size the larger domain produced 949,913 cells, well past the Student cap.

4. Solver configuration

Setting	Value	Setting	Value
Solver	Density-based, steady	Flux type	Roe-FDS
Turbulence	k-ω SST	Discretisation	First-order upwind
Density	Ideal gas	Wall BC	No-slip, isothermal 300 K
Viscosity	Sutherland (3-coef)	Residual criteria	1×10^{-4}

Courant number ramped down as Mach increased — 0.5 at Mach 5, 0.3 at Mach 7.5, 0.2 at Mach 10 — each case restarted from the previous converged solution. Second-order upwind destabilised the solution and first order was retained.

Case	Alt (km)	Mach	P∞ (Pa)	T∞ (K)	ρ∞ (kg/m³)	V∞ (m/s)	q∞ (Pa)
1	40	5.0	287.14	250.35	3.996e−3	1586	5025
2	45	7.5	149.10	264.16	1.966e−3	2444	5871
3	50	10.0	79.78	270.65	1.027e−3	3298	5585
—	60	15.0	21.96	247.02	3.097e−4	4726	3459
—	70	20.0	5.221	219.59	8.283e−5	5941	1462
—	75	25.0	2.388	208.40	3.992e−5	7235	1045

Cases 1–3 run in CFD. Mach 15–25 covered analytically — see §9.

5. CFD validation against theory

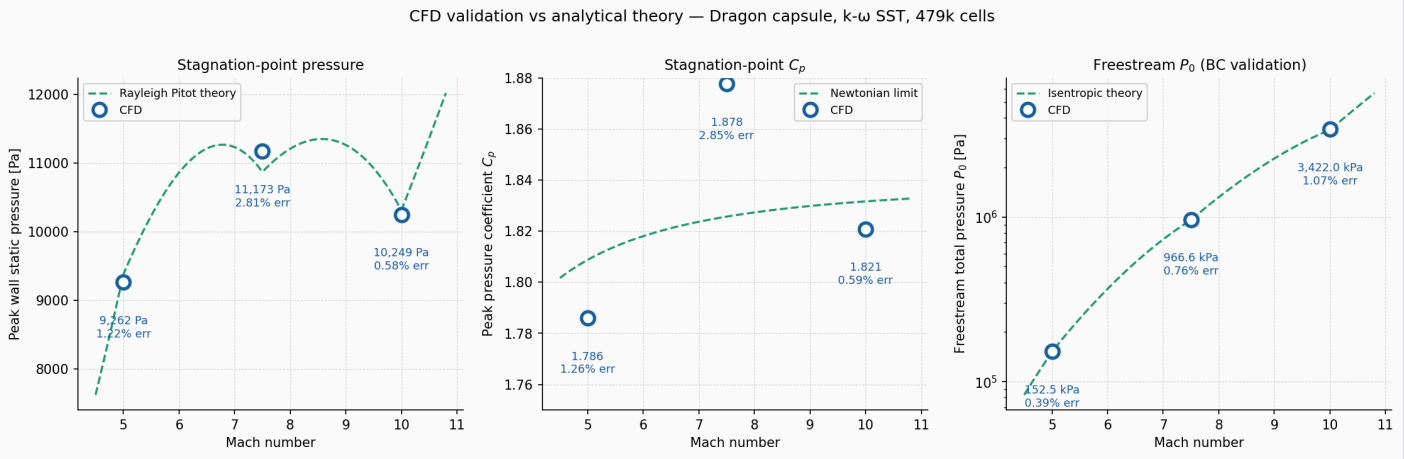


Figure 3 — Three independent checks: peak wall static pressure against Rayleigh Pitot, peak C_p against modified Newtonian, and freestream total pressure against isentropic theory. Maximum error 2.85%.

Mach	Wall P, CFD	Theory	Err	C_p CFD	Theory	Err	P_0 CFD	Theory	Err
5.0	9,262.0	9,376.1	**1.22%**	1.786	1.809	**1.26%**	152,514	151,923	**0.39%**
7.5	11,172.6	10,867.5	**2.81%**	1.878	1.826	**2.85%**	966,595	959,299	**0.76%**
10.0	10,248.8	10,308.9	**0.58%**	1.821	1.832	**0.59%**	3,421,984	3,385,802	**1.07%**

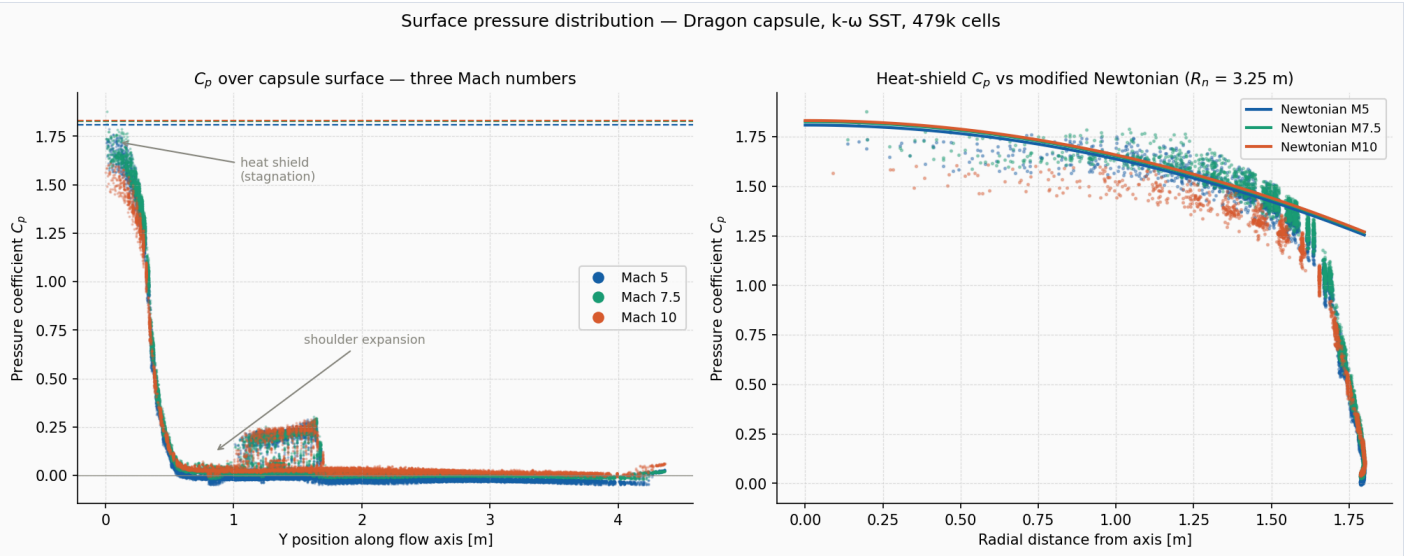


Figure 4 — C_p is nearly Mach-independent on the heat shield: theory predicts 1.809 at Mach 5 and 1.832 at Mach 10, under 2% apart across a doubling of speed, and the CFD reproduces that collapse. The sharp drop near Y ≈ 0.4 m is the shoulder expansion.

Three observations

- **The Mach 7.5 C_p exceeds its own theoretical bound.** At 1.878 it sits above the Newtonian limit of 1.826, which is not physically possible for a steady stagnation point. The cause is a single-facet pressure spike from shock smearing on a coarse grid — an artifact of taking a facet maximum. Area-weighted averaging is the appropriate cross-check.

- **Total pressure loss across the bow shock is severe.** At Mach 5, freestream P_0 of 152,514 Pa arrives at the wall as 9,262 Pa — a 93.9% loss, consistent with strong normal-shock behaviour.
- **Boundary conditions are validated separately from the solution.** Checking freestream P_0 against isentropic theory confirms the inlet delivers the intended conditions — a different question from whether the flow field is right.

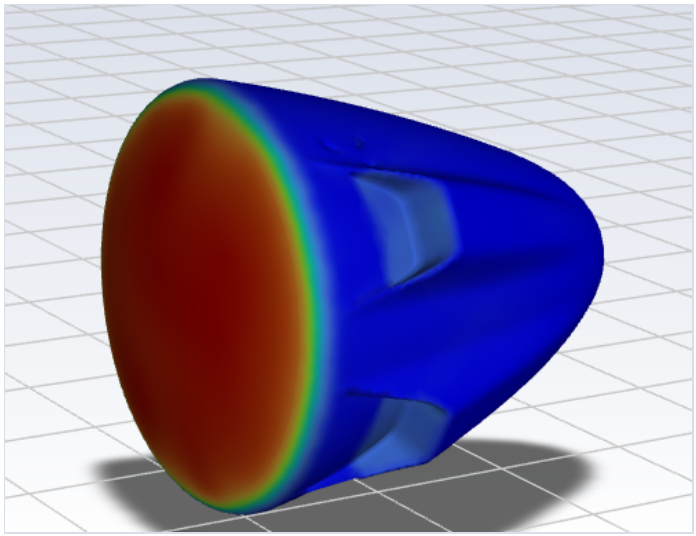


Figure 5 — Wall static pressure. Peak lands on the heat shield, not the nose: the check that caught a reversed flow-direction error.

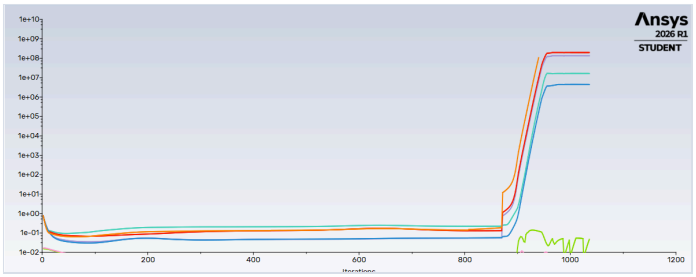


Figure 6 — Mach 15, cold start at Courant 0.25. Residuals idle near 10^{-1} for 750 iterations then blow up to 10^{+8}.

6. Stagnation-point heating

Fay & Riddell (1958), equilibrium boundary layer, using the CFD-fitted R_n = 3.248 m, a 300 K wall and Sutherland viscosity:

$$q_w = 0.763 \cdot Pr^{-0.6} \cdot (\rho_e \mu_e)^{0.4} \cdot (\rho_w \mu_w)^{0.1} \cdot \sqrt{(du_e/dx)} \cdot (h_{0e} - h_w)$$

Mach	Alt (km)	V (m/s)	T_0 (K)	Fay-Riddell	Sutton-Graves	Δ
5.0	40	1,586	1,502	2.15 W/cm²	2.44	11.8%
7.5	45	2,444	3,236	5.98	6.25	4.3%
10.0	50	3,298	5,684	11.07	11.11	0.3%
15.0	60	4,726	11,363	18.61	17.95	3.7%
20.0	70	5,941	17,787	19.59	18.44	6.2%
25.0	75	7,235	26,258	25.09	23.12	8.5%

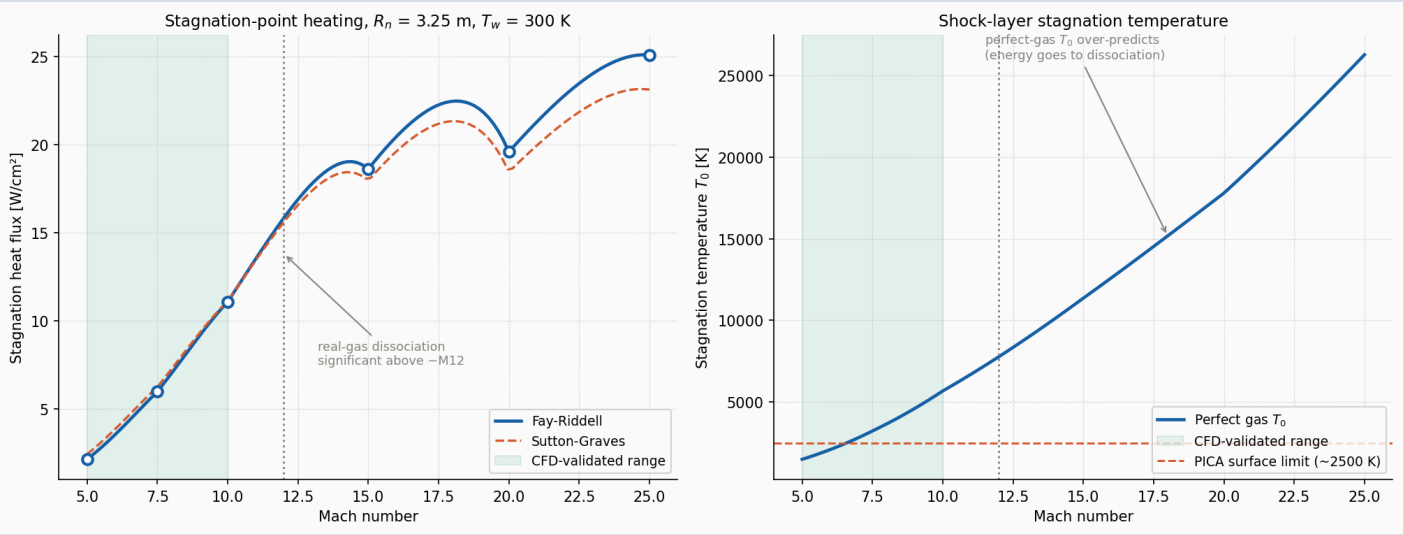


Figure 7 — Heating and shock-layer temperature, Mach 5–25. Two independent correlations agreeing to 0.3% at Mach 10 is what validates the implementation.

Why a correlation instead of CFD wall heat flux? Inflation layers could not be generated on this geometry, so the near-wall boundary layer is unresolved and solver wall heat flux is not trustworthy. Surface pressure is driven by the inviscid outer flow and remains valid — which is why pressure comes from CFD and heating is computed analytically. Anchoring a cheap correlation to high-fidelity data where high-fidelity is affordable, then extrapolating, is standard preliminary-design practice.

7. Entry trajectory and TPS sizing

Allen-Eggers ballistic entry through an exponential atmosphere. Entry velocity 7,600 m/s, flight path angle −5.5°, mass 9,500 kg, C_D 1.4, reference area 10.18 m², ballistic coefficient β = 667 kg/m². Peak Mach 27.7; 354 s from entry interface to 20 km.

TPS model: 1-D transient conduction, implicit backward Euler, 120 nodes. Front face receives $q_{conv} - \varepsilon\sigma(T^4 - T_\infty^4)$ with surface temperature capped at the material ablation temperature and excess flux booked as absorbed by ablation. Back face adiabatic — conservative, allowing no heat rejection into structure. Thickness sized to hold the bondline below 523 K.

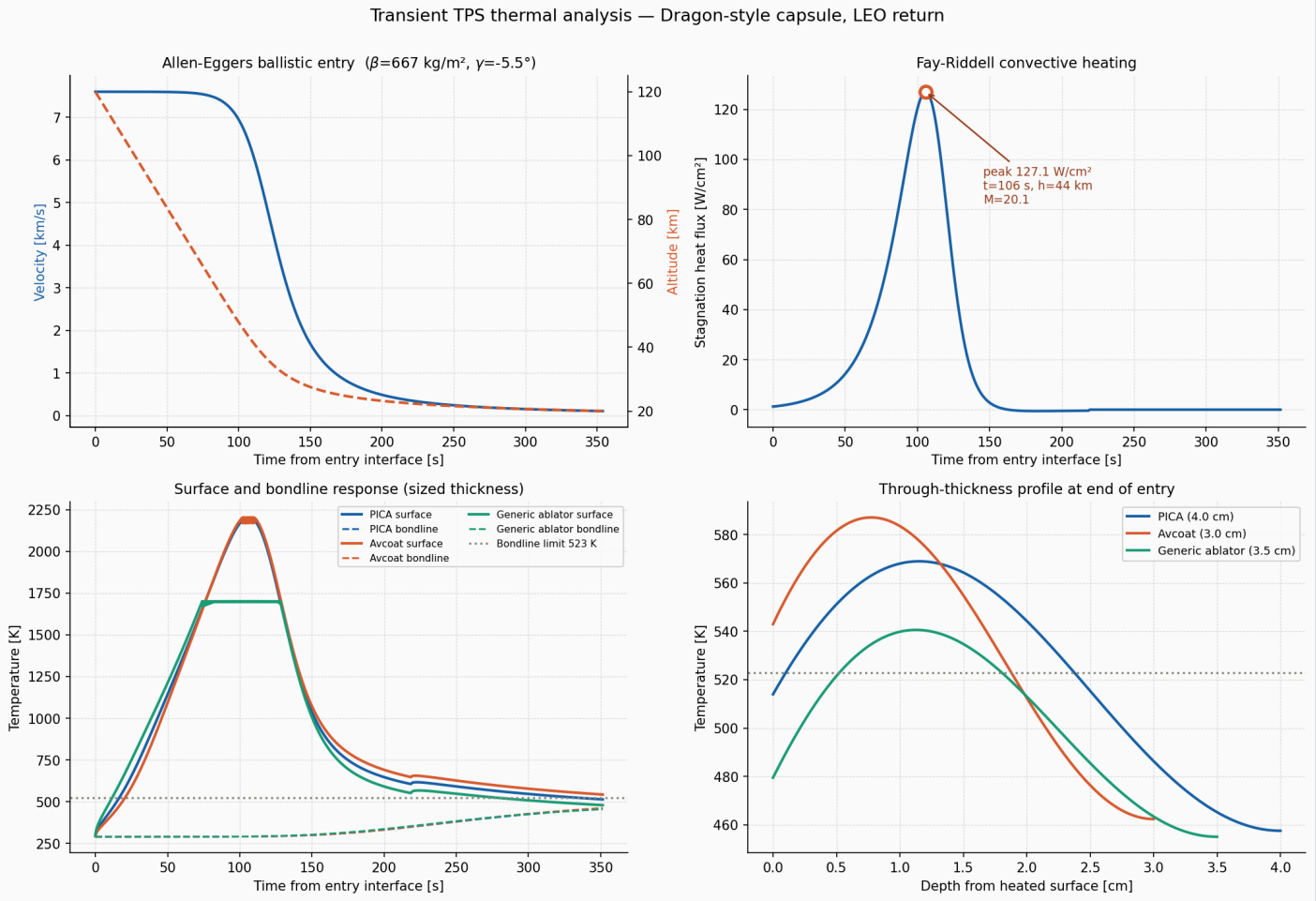


Figure 8 — Left to right, top to bottom: Allen-Eggers velocity and altitude history; Fay-Riddell heat flux peaking at 127.1 W/cm² at t = 106 s, 44 km, Mach 20.1; surface and bondline temperature response at sized thickness; through-thickness profile at end of entry.

Material	ρ (kg/m³)	k (W/m·K)	c_p	T_{abl} (K)	Thickness	Peak surface T	Peak bondline T	Areal mass
PICA	270	0.35	1,590	2,900	4.0 cm	2,203 K	**458 K**	**10.8 kg/m²**
Avcoat	512	0.35	1,465	2,200	3.0 cm	2,200 K	**462 K**	15.4 kg/m²
Generic ablator	264	0.213	1,255	1,700	3.5 cm	1,700 K	**455 K**	9.2 kg/m²

Why Avcoat needs less thickness but weighs more. Its lower ablation temperature caps surface temperature sooner, reducing the gradient driving heat inward — 3.0 cm against PICA's 4.0 cm. But at 512 kg/m³ it is nearly twice

as dense, so it still carries a **43% areal-mass penalty**. That trade is why low-density ablators displaced Avcoat-class materials on modern vehicles. The generic silicone ablator shows the lowest areal mass, but its 1,700 K limit means it would recede fastest in service; the model does not track recession, so that comparison understates its true thickness requirement.

> **These are preliminary sizing values, not design values.** They under-predict flight hardware by roughly a factor of two — actual PICA-X on Dragon is substantially thicker than the 4.0 cm computed here. The model captures the correct physics and the correct relative ranking between materials. No design margins have been applied.

8. Errors found and corrected

#	Issue	How it was detected	Resolution
1	Vehicle flying nose-first instead of heat-shield-first	Compared expected stagnation location against geometry	Flow direction set to (0, 1, 0)
2	Reference values left at solver defaults (ρ = 1.225, V = 1)	Exported C_p read 77–15,121 against a physical max of ≈1.8	Compute-from set to far-field zone; earlier data rescaled
3	Boolean subtract silently returned a no-op	Domain volume exactly matched a bare sphere	Switched to the Enclosure tool
4	Capsule retained as a solid cell zone	Cell Zone Conditions listed fluid + solid, shadow walls, contact interfaces	Suppressed the capsule body for physics
5	Node-based rather than cell-based C_p export	Mach 10 peak C_p read 1.611 against 1.821 from the facet maximum	Specified cell-centred export
6	Mesh exceeded the Student cell cap	949,913 cells at 4 m element size	Element size tuned to 6.5 m → 479,475 cells

> **Item 1 was the consequential one.** Stagnation heat flux scales as $1/\sqrt{R_n}$, so flying the low-radius nose into the flow rather than the large-radius heat shield would have substantially over-predicted heating and inverted the entire pressure distribution — while still converging cleanly and producing contours that looked entirely reasonable.

9. Limitations

- **No inflation layers.** Inflation generated inverted cells across multiple attempts; root cause is complex imported surface topology. The near-wall boundary layer is unresolved, so CFD wall heat flux is not used anywhere in this work. Surface pressure, driven by the inviscid outer flow, remains valid.
- **Mach 15 did not converge.** Two attempts: cold start at Courant 0.25 diverged to 10^{+8} with the stagnation monitor at 350 MPa against 6.4 kPa theory; Courant 0.05 was stable but residuals bottomed at iteration ~350 and climbed, settling 14% low. Shock thickness scales inversely with Mach, so a shock marginally resolved at Mach 5 is under-resolved at Mach 15, and the 512k cap prevents the refinement that would fix it. Mach 15–25 are covered analytically.
- **Calorically perfect gas.** γ = 1.4 throughout. Above ≈Mach 12, O_2 and N_2 dissociation becomes significant and effective γ falls well below 1.4. Perfect-gas theory predicts T_0 = 26,258 K at Mach 25; real shock-layer temperature would be nearer 8,000–10,000 K. This is why CFD above Mach 12 was not attempted.
- **First-order discretisation.** Second-order upwind destabilised the solution with a residual spike at iteration 3,000 on the Mach 5 case. Expected consequence is additional numerical diffusion and a slightly smeared shock.

- **Simplified ablation.** A temperature cap with energy bookkeeping, not a mass-loss-coupled model. No surface recession. Constant properties — no temperature dependence, pyrolysis or char layer. Radiative heating omitted. Stagnation point only.